



From Passive to Metacognitive: The Role of Mode of Engagement in Enabling Lasting Learning Using Digital Media

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Abstract

The current study presents a theory-driven, person-centered analysis of the relationship between different modes of cognitive and metacognitive engagement and immediate and delayed knowledge in chemistry. In a classroom study with high ecological validity, we inferred cognitive and metacognitive engagement from log files of secondary school students' ($N=235$) activities completed within the context of a digital chemistry textbook. We used the ICAP framework to classify cognitive engagement as either passive, active, or constructive, and we extend the framework to include metacognitive engagement. A latent profile analysis identified six distinct latent profiles, characterized by different distributions of student cognitive and metacognitive engagement while learning with the digital textbook: Disengaged; Low Metacognition; Disfavors Active; Moderate Engagement; Metacognitive, but not Constructive; and High Overall Engagement. We found that these profiles were predictive of immediate (one week later) and delayed (six to eight weeks later) chemistry knowledge, even after controlling for prior knowledge. Post-hoc tests indicated that “High Overall Engagement” and “Metacognitive, but not Constructive” were the best performing profiles in both immediate and delayed learning outcomes. Interestingly, profile membership was not predicted by self-reported chemistry self-concept or interest, measured prior to learning with the digital textbook. Our find-

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Educational Relevance Statement The present study analyzed which activities best predicted learning outcomes in a digital chemistry textbook. We identified six subgroups of learners, defined by the different patterns of engagement frequency with the digital textbook. Our findings showed that constructive activities (e.g., activities where students were asked to create new content by writing) and metacognitive activities (e.g., activities that ask students to monitor their current knowledge) were related to lasting learning. We propose that metacognitive engagement is an important extension of existing cognitive engagement frameworks (ICAP), specifically during individual, self-regulated learning.

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ings provide a novel extension of the ICAP framework and highlight the role of metacognitive engagement for lasting learning in digital learning environments.

Keywords Digital Learning · Science Education · ICAP framework · Metacognition · Cognition

A critical question regarding any curriculum is “Are the learning activities that we are having our students perform effective?” With analog media, it is often impossible to answer these questions, as there is usually no means of collecting data from students concurrently when they are interacting with a specific element of a curriculum without interfering with the actual learning process. Digital resources, on the other hand, have become increasingly used in educational practice (OECD, 2023; Weller, 2022) and allow for the unobtrusive collection of fine-grained usage data (Lerche & Kiel, 2018). These log data can measure students’ usage, engagement, and performance with regard to different learning activities in regular classroom practice, without the need for artificial laboratory learning environments. Furthermore, students can take non-linear, self-regulated paths through digital learning environments, allowing for theoretically interesting variability in students’ engagement with these resources (Azevedo et al., 2019; Winne, 2022). The interactive-constructive-active-passive (ICAP) framework (Chi & Wylie, 2014) is frequently used for classifying students’ cognitive engagement (e.g., Wong et al., 2024). However, the ICAP framework focuses only on cognitive engagement and does not explicitly consider the role of self-regulated learning and metacognitive engagement, which is particularly relevant during self-paced technology-enhanced learning (Brady et al., 2022). In the present study, we investigate whether metacognitive and cognitive engagement predict secondary school students’ immediate and delayed chemistry learning outcomes, beyond prior knowledge. We provide an ecologically valid (albeit correlational) test of the ICAP framework and suggest an extension with metacognitive engagement. Using a theory-driven analysis of log file data collected during regular chemistry lessons, we aim to answer the following research questions:

RQ1. What student profiles can we identify based on the type of engagement (passive, active, constructive, metacognitive) within a digital learning environment?

RQ2. Can profile membership predict immediate and delayed knowledge above and beyond prior knowledge?

RQ3. Which types of engagement are related to higher immediate and delayed knowledge?

Theoretical Background

Student Engagement in Digital Learning Environments

Digital learning environments enable high amounts of learner control due to their flexible and self-paced nature (Scheiter & Gerjets, 2007; Zhang et al., 2020). At the

same time, the lack of external regulation by a teacher and the open-ended nature of digital learning environments poses additional demands on students to self-regulate their learning (Lajoie & Azevedo, 2006; Scheiter & Gerjets, 2007). Learning in non-linear digital learning environments relies on students' autonomy and accurate monitoring while progressing through the learning units. Students' engagement (Littlejohn et al., 2016; Vytasek et al., 2020) and learning success (Lin, 2022; Winters et al., 2008) in computer-based learning environments can depend on their ability to self-regulate their learning and overcome distractions (Brady et al., 2022; Scheiter & Gerjets, 2007). In particular, learning environments that provide students with control over the sequence and the type of learning materials to be studied, are more effective for students with higher metacognitive and cognitive abilities (Scheiter et al., 2009; Winters et al., 2008).

In the case of the eChemBook learning platform, under investigation in this paper, students can skip around the learning content, rewind videos, re-engage with simulations, write summaries, and solve drawing exercises to better understand the learning material. Furthermore, the eChemBook gives students the opportunity to make metacognitive judgments for each learning unit and to review learning material if they judged their learning as incomplete. Thus, the digital nature of this learning platform allows for more individual variability in student engagement beyond what is traditionally seen with teacher-led instruction or analog learning materials and provides students the opportunity to learn the material in a self-regulated manner.

Accordingly, digital environments are uniquely suited to capturing indicators of students' self-regulated learning and engagement in vivo (Carvalho et al., 2016; Viberg et al., 2020). This flexibility has been taken advantage of by numerous researchers at the intersection of learning analytics and self-regulated learning (Roll & Winne, 2015), both in observational (e.g., Kizilcec et al., 2017) and intervention studies (e.g., Dietrich et al., 2021). But, the breadth of data available in digital environments can be both a blessing and a curse for researchers. Given the fine-grained nature of log data, it is often difficult to analyze in a theoretically informed manner (Alhazbi et al., 2024; Marzouk et al., 2016).

Cognitive Engagement

We adopted the ICAP framework (Chi, 2009; Chi & Wylie, 2014) as a means of classifying learning activities into different modes of cognitive engagement. The ICAP framework offers a taxonomy that categorizes learning activities into four modes of cognitive engagement — passive, active, constructive, and interactive. Specifically, the ICAP framework derives its name from the hypothesis that **interactive** activities will be more effective than **constructive** activities, **active** activities, and **passive** activities in that order ($I > C > A > P$). Interactive learning activities require students to engage with one another (e.g., peer instruction, debating). In other words, under the ICAP taxonomy, interactive activities always involve at least two students. Constructive activities require students to go beyond the information presented to them in the class and generate new content (e.g., writing discussion posts, creating concept maps). Active learning activities require manipulation of the learning content while engaging in learning activities (e.g., note taking, working with a simulation), while

passive activities refer to receiving information without additional engagement (e.g., listening to a lecture without taking notes).

In prior research, the ICAP framework has been used to classify different types of learning activities during technology-enhanced learning based on self-reports by teachers (Antonietti et al., 2023; Wekerle et al., 2024) and students (Wong et al., 2024; Zepeda & Nokes-Malach, 2021). Other studies have used videotapes of classroom observations to classify students' engagement based on the ICAP framework (Goldberg et al., 2021; Hsiao et al., 2022). These studies demonstrate the practical application of the ICAP framework for distinguishing between different levels of (intended) cognitive engagement in different learning tasks using different measurements. In the present study, we use the ICAP framework as a means to categorize the different types of learning activities within a digital learning environment based on logfiles of students' activities. Ultimately, we acknowledge that no method of categorizing disparate learning activities will be fool-proof, but in the context of the present analyses, the ICAP framework balances theoretical precision with pragmatic constraints on analytic complexity.

As the eChemBook is meant for students to use individually, we focus on the ICAP modes relevant for individual learning: passive, active, and constructive. Furthermore, we use multiple different types of activities within each ICAP mode. Due to the importance of self-regulated learning during individual learning with an open-ended digital learning environment (Lajoie & Azevedo, 2006), we additionally investigate the role of metacognitive engagement. Although many studies have investigated students' levels of cognitive engagement in relation to cognitive and motivational variables (e.g., Wekerle et al., 2024), the role of self-regulated learning has rarely been discussed in relation to the ICAP framework (cf. Zepeda & Nokes-Malach, 2021).

Metacognitive Engagement

The ability to self-regulate one's learning is positively related to academic achievement (Dent & Koenka, 2016) and is particularly important for successful learning in computer-supported learning environments (Azevedo et al., 2010). Self-regulated learning involves the monitoring and regulation of cognition, affect, and behavior before, during, and after learning (Azevedo et al., 2017). Different models of self-regulated learning define metacognition as a central aspect of self-regulated learning (e.g., Pintrich, 2000; Winne & Hadwin, 1998; Zimmerman, 2000). Under Schraw et al.'s (2006) framework, metacognition has two dimensions — metacognitive knowledge and metacognitive regulation. Metacognitive knowledge includes declarative knowledge (knowledge of own cognitive abilities), procedural knowledge (knowledge about different learning strategies), and conditional knowledge (knowledge about when to use specific learning strategies). Metacognitive regulation refers to the monitoring and control of cognitive processes during learning (Nelson & Narens, 1990). During monitoring, learners check their understanding of the learning materials and if they judge their understanding to be incomplete, they engage in control processes by changing their behavior or cognitive processes (e.g., re-reading text or changing the learning strategy) to facilitate their understanding (Nelson & Narens, 1990). Monitoring interventions are particularly effective for students' achievement

and motivation (Dignath et al., 2023). In particular, monitoring tools that ask students to monitor their understanding of the learning content are highly effective (Dignath et al., 2023). Therefore, when investigating learning in open-ended learning environments, it is important to not only consider students' use of cognitive strategies, but their monitoring and regulation as well.

Although the ICAP framework does not explicitly consider the role of metacognition for learning, there is extensive research literature on the potential benefits of metacognitive engagement on both memory (Double & Birney, 2019; Double et al., 2018) and learner-controlled decisions (Dent & Koenka, 2016; Lyons & Zelazo, 2011). For example, Janes et al. (2018) suggest that metacognitive engagement helps students adopt learning goals that foster greater learning. That is, by being prompted to consider their long-term learning when answering metacognitive prompts, learners may be more likely to adopt more effective study strategies. This is also in line with the literature on study-time allocation (Son & Metcalfe, 2000; Tekin, 2022), which suggests that metacognitive judgments may be used by students to identify the content that is most at risk of being forgotten or not sufficiently well understood (see Ackerman, 2014; Thiede & Dunlosky, 1999). Then, in turn, this at-risk-of-being-forgotten material is assigned more study time, which should in turn (all things equal) increase the chance of knowledge retention at a delay. Altogether, metacognitive monitoring seems to improve learning through direct mnemonic benefits, increased attention to goal-setting, and better regulation of study-time allocation. Despite its importance for effective learning, metacognitive engagement, in particular monitoring, is not considered by the ICAP framework. In the present study, we draw upon theories of self-regulated learning and metacognition (Nelson & Narens, 1990) and propose metacognitive engagement as an additional mode of engagement (see Fig. 1), particularly relevant for students engaging with a learning environment that provides a variety of cognitive and metacognitive activities like the eChemBook. For example, metacognitive activities that foster students' monitoring might influence the following selection of tasks and the subsequent choices of different modes of cognitive engagement.

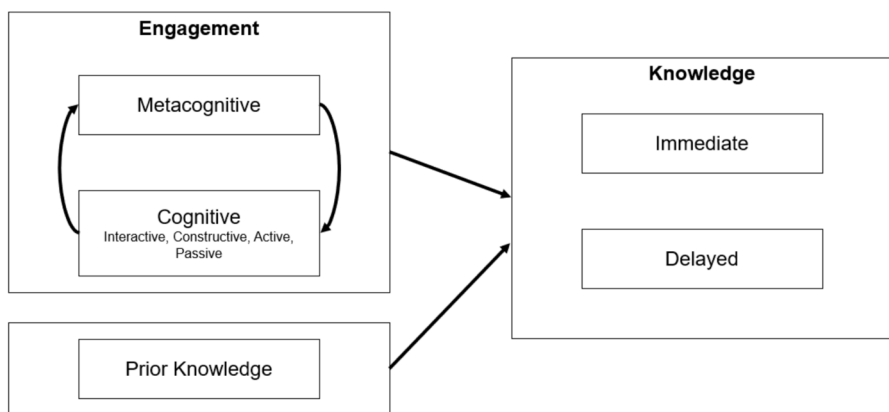


Fig. 1 Conceptual Model of the Relationships Between the Modes of Engagement and Knowledge

A Person-Centered Approach for Modeling Individual Differences in Engagement

Person-centered approaches have gained increasing adoption in applied psychology and education research (e.g., Hong et al., 2020; Howard & Hoffman, 2018; Muenks et al., 2024; Schnitzler et al., 2021; Wong et al., 2024) due to their flexibility in modeling multiple indicators. Unlike variable-centered approaches (e.g., hierarchical linear modeling, structural equation modeling), which estimate mean changes in outcomes while controlling for other variables, person-centered methods identify subpopulations within a sample, enabling the examination of outcomes that emerge from distinct constellations of potentially interacting behaviors (Howard & Hoffman, 2018). In this study, we use latent profile analysis (LPA), a person-centered technique that classifies individuals into latent profiles based on continuous predictor variables and a preselected number of profiles. Latent profile analysis can be seen as an alternative to structural equation modeling, where the goal is to create categorical (as opposed to continuous) latent variables (Van Lissa et al., 2023). The algorithm determines the best-fitting profiles within the data, assigning each participant a probability of categorical profile membership. Typically, researchers test varying numbers of profiles in conjunction with calculating model fit statistics to identify the best fitting number of latent profiles to describe the patterns in the data.

Previous research has used LPA to predict academic achievement based on profiles of motivational beliefs (e.g., Hong et al., 2020; Muenks et al., 2024), behavioral and self-reported engagement (e.g., Schnitzler, et al. 2021), and, most relevantly, Wong et al. (2024) performed a survey aimed at capturing students' ICAP engagement. Wong et al.'s study measured ICAP engagement in an engineering class lab exercise through items such as "The use of the [learning tool] helped me discuss with a peer about the [...] concepts more than I could with a lecture." (p. 326). Wong et al. identified five profiles, ranging primarily in level of engagement (from Disengaged to Fluctuating Engagement). Wong et al. noted that the Fluctuating Engagement profile was notable in that it represented students who heavily used all modes of engagement defined by the ICAP framework. Wong et al. interpreted the existence of this profile as deviating from what one would expect, given the lack of support for passive learning under the ICAP model. Overall, Wong et al. found that change in academic performance from the pre-test could be predicted by their five ICAP profiles. However, we note that there is room to (partially) replicate and expand upon the work of Wong et al. by measuring, identifying, and associating ICAP-related learning behaviors (as opposed to self-report measures) to learning outcomes in scientific classes.

The Present Study

The present study uses the ICAP framework as a means of theoretical organization to evaluate the potential benefits of different learning activities that students engage in while learning with a digital chemistry textbook, known as *eChemBook* (Richter et al., 2018; Ulrich, 2019). The *eChemBook* learning environment is designed to teach students about the particle model of matter using a variety of learning tasks, which afford different modes of cognitive and metacognitive engagement. Some of these tasks are passive (e.g., reading or watching videos), active (e.g., navigating playback

of videos, using a simulation), constructive (e.g., writing summaries, drawing), or metacognitive (e.g., making judgments of learning). This tool is notable for its wide variety of learning activities and the self-regulated nature through which students can choose to engage with these activities. Given the variety of ways one can engage with the *eChemBook*, the present study investigates whether cognitive and metacognitive engagement predict students' chemistry knowledge, after controlling for prior knowledge. Furthermore, we propose and test an extension of the ICAP framework that includes metacognitive engagement as an additional mode of engagement).

Methods

Transparency and Openness

We report how we determined our sample size, all data exclusions, all manipulations, and all measures in the study. Although raw data cannot be shared openly due to data collection agreements, full analysis code and output can be found at the following OSF link: https://osf.io/uypfk/?view_only=11d819491f7b4b158ef984ebf02f470e. All models were fit with the R version 4.4.2 (R Core Team, 2024). The library *tidyLPA*, version 1.1.0 (Rosenberg et al., 2018) was used to estimate latent profiles using the *mclust* algorithm. The library *lavaan*, version 0.6–17 (Rosseel, 2012), was used to fit SEM models. This study design and analyses were not pre-registered. A fixed seed was set prior to estimating the latent profiles. Missing data in variables used for the LPA analysis were imputed using multiple imputation via the *mice* package, version 3.17.0 (van Buuren & Groothuis-Oudshoorn, 2011) prior to estimating the LPA. Full reproducibility information, including random seeds and a full listing of packages used with version information, can be found at the end of the analysis output found on OSF.

Participants

All together 396 students participated in the *eChemBook* study across 15 different eighth-grade science classes. These 15 classes were drawn from two states within Germany. Eight classes ($n=206$) were recruited from a city within Lower Saxony. The remaining seven classes ($n=190$) were recruited from a city within Baden-Württemberg. The data collection process for the former eight classes has been previously described by Ulrich (2019). Data from the latter seven classes was previously described by Richter et al. (2018). In addition to the basic *eChemBook* content shared by the data collection sites, each of the study locations tested one of two potential efficacy-boosting interventions related to the design of the learning materials in the *eChemBook*. In Lower Saxony, researchers tested a low-interactivity versus high-interactivity intervention (see Ulrich, 2019). The term “interactivity” in Ulrich (2019) is related to interacting with the learning material and not to collaborating with others (i.e., distinct from the usage of “interactive engagement” in the ICAP framework). The intervention compared two versions of the drawing and matching tasks with varying levels of interactivity. The high interactivity tasks provided additional draw-

ing support or verification feedback, whereas the low interactivity tasks provided students with a model solution and no additional support. In Baden-Württemberg, Richter et al. (2018) compared two versions of the eChemBook, investigating the effects of basic versus extended signaling. The basic signaling condition included text signals (e.g., headings) and basic multimedia integration signals (labels linking texts and pictures). The extended signaling condition included additional multimedia integration signals (e.g., color coding) to additionally support the integration of information from text and pictures (Richter et al., 2018).

For the present analyses, we included only participants who were in the high-interactivity condition, as the drawing and matching tasks were the same as in both signaling conditions. We did, however, retain students in both signaling conditions, as students in both conditions could engage in the same activities. In our final sample, most participants (64%) were in the extended signaling condition (150 out of 235).

Overall, we analyzed data from the 235 participants, from whom we were able to link pre-test, process, and post-test data. Of these participants, 49% were male ($n=116$), 37% were female ($n=87$), and 14% ($n=33$) did not indicate their gender. Participants were between 13 and 17 years old ($M=14.21$, $Mdn=14$, $SD=0.67$). Age data was missing for 36 participants (15%).

The research project received ethical approval from the Ministry of Education in Lower Saxony and the school administration in Baden-Württemberg. The students and their parents provided written informed consent before participating. Participation in the study was voluntary.

Materials

Structure of the eChemBook

The eChemBook is a digital chemistry textbook that was designed as part of a project funded by the German Research Foundation (Deutsche Forschungsgemeinschaft) as a collaboration between scientists, a textbook publisher, and an interactive whiteboard manufacturer (Richter et al., 2018; Ulrich, 2019). Students worked on six learning units on the topic of the particle model of matter. All units have the same five structural elements: a short motivational teaser, a basic text providing students with background information, a video experiment, two learning activities, and a summary page. An overview of the number of pages and the number and type of multimedia materials (e.g., video, simulations) within each unit is provided in Table 4. Although the units shared this structure, students were not obligated to navigate the pages of the units in any predefined sequence but they could skip and revisit different pages if they wanted to. To illustrate each element, we will use examples from the fourth learning unit “Particles under pressure”.

Motivational Teaser

The introductory page provided the motivation for studying the topic with an overview of the learning unit and a reference of the topic to students’ daily lives. For

example, in the fourth learning unit, the motivational teaser used pumping a bicycle tire as a reference.

Basic Text

The basic text provided students with the necessary background information to work with the video experiments and the various learning tasks contained within the digital textbook. The basic text contained between 3 and 7 pages with text, images, animations, and simulations. In the fourth learning unit, the first two pages provided text explanations about the role of the number of particles, energy, and volume. On the next page, students could investigate their relationships with a simulation (see Fig. 2).

Next, students watched an animation visualizing the influence of volume on pressure using the bicycle pump example. On the last page, students investigated how volume, number of particles, and energy affect pressure with a simulation (see Fig. 3). Next to the simulation, students solved a cloze (“fill-in-the-blank”) test, in which they chose one of two options that correctly explained the relationships they observed in the simulation. For example, “*If you increase the number of particles while maintaining the same energy and volume, the pressure decreases/increases.*” Students had to select whether “decreases” or “increases” was the correct answer.

Darstellung der Zusammenhänge zum Thema Druck

Du kennst nun schon die Einflussfaktoren Energie und Teilchenanzahl. Es gibt aber noch einen weiteren Faktor, um den Druck in einem Gefäß zu verändern: das Volumen. Bisher sind wir immer davon ausgegangen, dass das Gefäß sich nicht verändert. Das Gas hatte immer das gleiche Volumen, also den gleichen Rauminhalt, zur Verfügung. Wir können aber auch die Raumgröße verändern, während die Energie und die Teilchenanzahl gleich bleiben. Das Video auf der nächsten Seite zeigt dir den Einfluss des Volumens auf den Druck am Beispiel einer Fahrradpumpe.

Überlege dir Alltagsbeispiele für jeden Einflussfaktor.

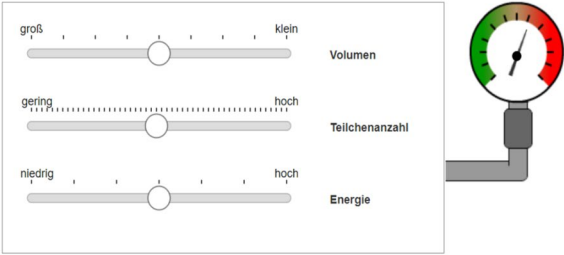


Fig. 2 An Example of a Basic Text Page Including Text and a Simulation

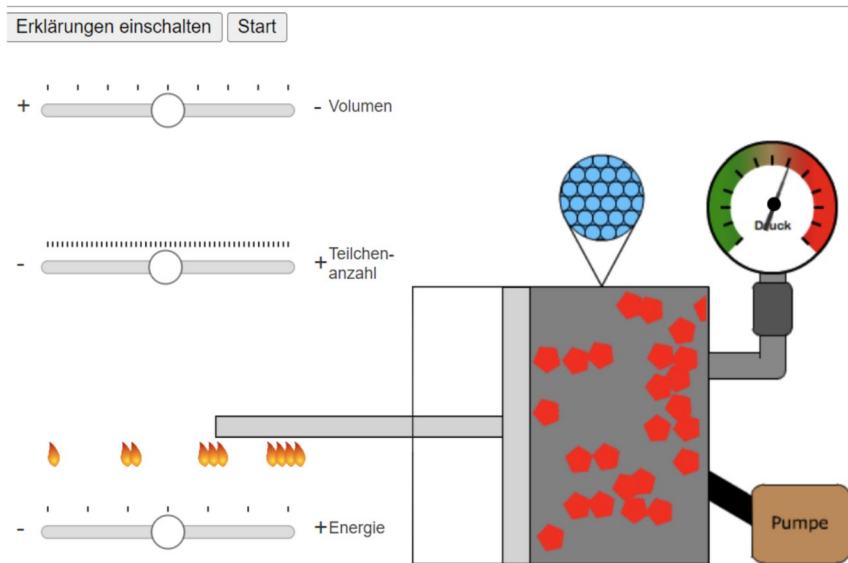


Fig. 3 Screenshot of Bicycle Pump Simulation

Video Experiment

The video experiment contained two pages. On the first page, the experimental setup was described and instructions on how to conduct the experiment were provided. Students did not perform the experiments themselves but watched a video of the experiment. On the second page, a video of the experiment was provided and students wrote down their observations and interpreted the experiment results in two text boxes.

Drawing and Matching Exercises

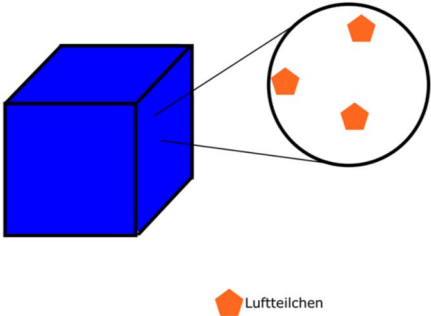
Each unit included two tasks in which students drew particle-level representations of a phenomenon and provided a written explanation. In the fourth unit, students solved two exercises involving drawing depictions of scientific concepts. One exercise asked the following from students: *“There is only air in the box. Draw in the circle what you think the inside of the box looks like at the particle level. Explain your drawing.”* See Fig. 4, for a visualization of the exercise and an example solution (*“The air in the box consists of particles. The distances between the particles are large. There is nothing between the particles.”*).

Summary

The summary page summarized the main concepts of the learning unit. Students had the option to select which concepts they could explain through a checkbox interface. For example, in the fourth unit, students could select checkboxes associated with two

Aufgabe 2: Was befindet sich zwischen den Teilchen? - Musterlösung

In der Kiste befindet sich nur Luft. Zeichne in den Kreis, wie du dir das Innere der Kiste auf Teilchenebene vorstellst.



Die Luft in der Kiste besteht aus Teilchen. Die Abstände zwischen den Teilchen sind groß. Es gibt nichts zwischen den Teilchen.

Du kannst dieses Teilchen verwenden, um die Luft in der Kiste darzustellen.

Zurück

Fig. 4 Example Solution to a Drawing Task

concepts: pressure and factors influencing pressure. In addition to the checkboxes associated with the learning concepts, students could, but were not required to, also indicate on a traffic light scale how well they could explain the content of the entire learning unit: red (not at all), yellow (a little), and green (very well), see Fig. 5. The traffic light scales and checkboxes aimed to foster metacognitive monitoring by giving students the opportunity to make judgments of learning for each learning unit. Students could choose whether to provide these judgments. After completing a unit, students could write a summary of the main concepts and any open questions they had in the learning map (i.e., main menu) of the eChemBook. In the learning map, all learning units were shown and students could see their summaries and their scores on the traffic light scale for the units they already provided a score. As such, students could get an overview of which units they should further review.

Measures

Process Measures

Students' engagement with the different types of materials within the eChemBook were automatically recorded as participant-level log files. Each activity was logged for every student, and the sum or the frequency of activities (e.g., the number of videos played, frequency of using the traffic lights) was used as an indicator of engagement with each relevant activity. Each type of activity was then classified based on the ICAP framework to reflect the mode of cognitive (passive, active, constructive) and metacognitive engagement, forming four categories, see Table 1. These activities were used as indicators for the modes of cognitive and metacognitive engagement in the analyses. There were three passive activities, four active and constructive activities each, and two metacognitive activities.

Welche Konzepte dieser Lerneinheit kannst du erklären?

☒ Druck

☐ Einflussfaktoren des Drucks

Wie gut kannst du die Inhalte dieser Lerneinheit erklären?

gar nicht

ein wenig

sehr gut



Fig. 5 Example Metacognitive Monitoring Measures Located on Each Summary Page

Knowledge Pre-Test

The pre-test comprised 15 verification items (Cronbach's $\alpha = .43$) that assessed students' prior domain-specific chemistry knowledge. Students judged each item as either true or false; if their judgment was correct, they received 1 point, resulting in a maximum score of 15. The items measured different conceptions students have regarding the particle model of matter using one or two items per conception (for a detailed description of the item construction, see Richter et al., 2018).

Table 1 Indicators of Cognitive and Metacognitive Engagement from Logfile Data

Mode	Activity	Description
Passive	Play videos	The sum of played videos, only pressing play
	Play animations	The sum of played animations, only pressing play
	Pages with static pictures	Time (in minutes) spent on pages with static pictures
Active	Interactions with animations and videos	The sum of interactions with animations and videos other than play (e.g., pausing or re-starting an animation, showing or hiding explanations)
	Interactions with simulations	The sum of interactions with the simulations (e.g., changing variables, showing or hiding explanations)
	Matching tasks	The sum of edited matching tasks
	Cloze (fill-in-the-blank) test	The number of correct entries students provided in cloze (“fill-in-the-blank”) texts
Constructive	Texts in experiments	The sum of written interpretations in experiments
	Summaries written	The total number of summaries students wrote after finishing a learning unit
	Word count	The average word count across all written summaries and written interpretations in experiments
	Drawing tasks	The sum of edited drawing tasks
Metacognitive Monitoring	Traffic lights	At the end of each unit, students could provide a rating (1–3) of how well they could explain the content of the learning unit. The frequency of providing the traffic light ratings is used
	Concept checkboxes	At the end of each unit, students selected the number of concepts they thought they could explain. The number of concepts checked is used

Domain-Specific Academic Self-Concept

Domain-specific academic self-concept in chemistry was measured using 5 items (Größ-Niehaus, 2010) on a scale (Cronbach's $\alpha = .86$) ranging from 0 (*I do not agree at all*) to 3 (*I completely agree*). An example item is “*I can also solve difficult problems in chemistry.*”

Domain-Specific Interest

Domain-specific interest in science was measured using 5 items on a scale (Cronbach's $\alpha = .87$) ranging from 0 (*I do not agree at all*) to 3 (*I completely agree*). The items from Wilde et al. (2009) were adapted to the science context. An example item is “*I am interested in learning new things in science subjects.*”

Immediate and Delayed Knowledge Post-Test

The immediate post-test comprised the same 15 verification items (Cronbach's $\alpha = .65$, $\max_{\text{score}} = 15$) from the pre-test, an additional 29 multiple-choice items (Cronbach's $\alpha = .87$, $\max_{\text{score}} = 29$), and eight drawing items (Cronbach's $\alpha = .86$, $\max_{\text{score}} = 60$). To facilitate the comparison between the different tests, the percentage correct score was used. The items measured students' knowledge in three sub-categories: (mis)conceptions (verification items), recall (multiple-choice items), and transfer (drawing items). The three sub-categories were used as indicators for the latent variable immediate knowledge.

The items in the delayed post-test were identical to the items in the immediate post-test: 15 verification items (Cronbach's $\alpha = .71$), 29 multiple-choice items (Cronbach's $\alpha = .89$), and eight drawing items (Cronbach's $\alpha = .84$). The three sub-categories of the delayed test were again used as indicators for the latent variable delayed knowledge.

Procedure

The procedure was largely the same across both sites of data collection and included a pre-test session (45 min) during which students were introduced to the eChem-Book. Demographic information, academic self-concept, interest, and prior knowledge were assessed. After the pre-test session, students worked individually with the eChemBook during class in two learning sessions of 90 min each. The teacher and one of the researchers were present during the learning sessions resulting in a controlled but ecologically valid setting. Students worked on units 1 to 3 during the first learning session and on units 4 to 6 during the second learning session. The two learning sessions were between 5 to 7 days apart, depending on the chemistry class schedule of the different schools. The immediate post-test (45 min) was administered within one week after the second learning session (Richter et al., 2018; Ulrich, 2019). Students had no additional chemistry lessons before the immediate post-test. The delayed post-test (45 min) was administered between 6 and 8 weeks after the immediate test.

Analytic Approach

To test the potential effects of the eChemBook on students' acquisition of chemistry knowledge, we first analyzed log data reflecting students' actions within the online textbook using the *mclust* algorithm (Scrucca et al., 2023) in the *tidylpa* package (Rosenberg et al., 2018) to test latent profile analysis models using different numbers of profiles. We then constructed two primary SEM models of interest, using latent profile membership and prior knowledge as predictors of immediate and delayed knowledge. All SEM models were fit with the R library *lavaan*, version 0.6–17 (Rosseel, 2012) using full-information maximum likelihood estimation and accounting for clustering by class using cluster robust standard errors. As measures of model fit, we report the chi-square statistic (χ^2), the comparative fit index (CFI), the Tucker-Lewis index (TLI), the root-mean-square error or approximation (RMSEA), and the standardized root mean square residual (SRMR).

Results

Descriptive statistics of all variables used in the following analyses are shown in Table 5.

Latent Profile Analysis

Determining the Number of Profiles (RQ1)

We tested a series of models with increasing numbers of latent profiles, ranging from 1 to 10 profiles. A comparison of Bayesian Information Criteria along with a preference for parsimony suggested that a six-profile solution was the best solution among the tested models, see Fig. 6. Analysis of more models with larger numbers of profiles showed diminishing marginal returns to model fit (for a full table of model comparison indices including AIC and entropy, see Table 6 in the appendix). In these more complex models with greater numbers of profiles, we increasingly found that profiles that were defined by small sample sizes and resisted theoretical interpretation (see Bray & Dziak, 2018). The entropy value, which reflects the weighted average of the posterior classification probabilities (Collins & Lanza, 2013) of this six-profile solution was acceptably high (entropy = 0.98; Bauer, 2022). We also inspected the sizes of the individual profiles and their interpretability, to ensure that the six-profile solution was reasonable.

Interpreting the Profiles

Six distinct profiles were identified, see Figs. 7 and 8 for a detailed and summarized visualization. The first profile ($n=37$, 16%) was characterized by low values in all indicators; therefore, we called the profile *Disengaged*. The second profile ($n=15$, 6%) was characterized by very low values in both metacognitive indicators, average values in the constructive indicators, and slightly above average values in all

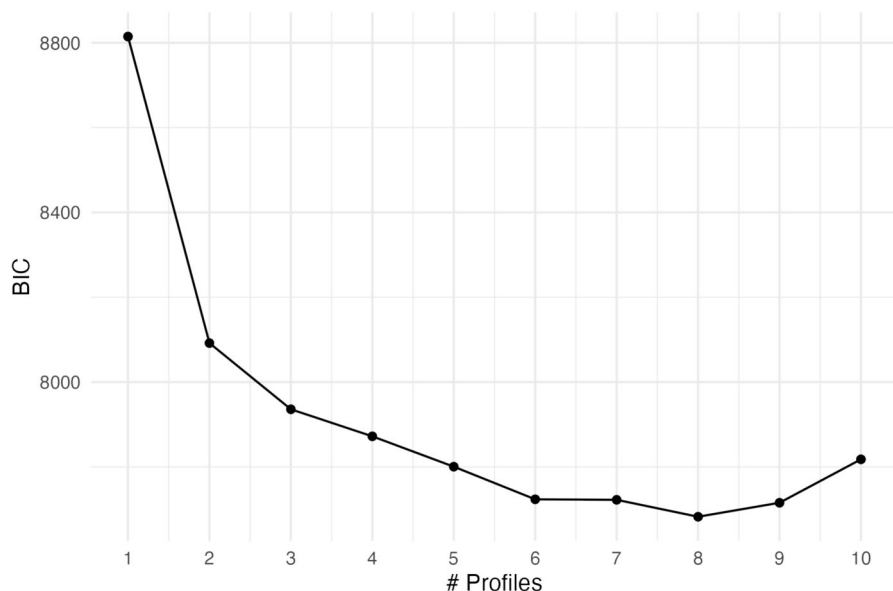


Fig. 6 Comparison of Latent Profile Models by BIC

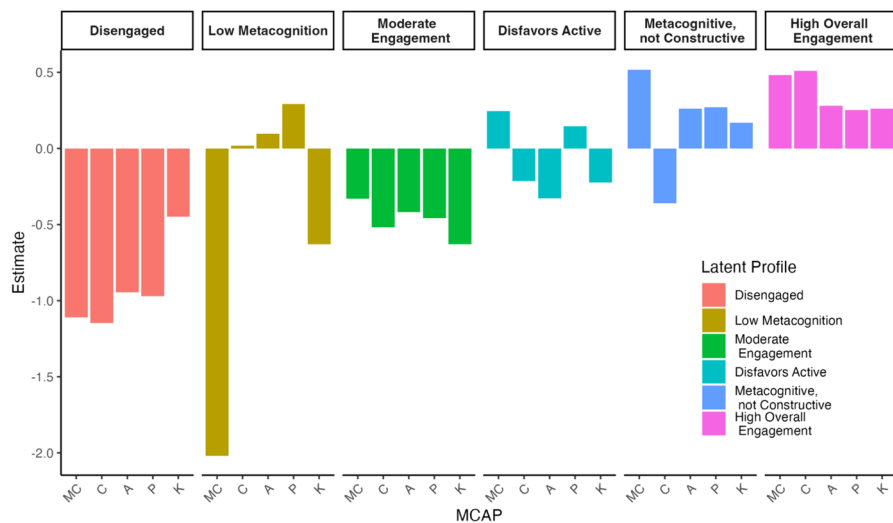


Fig. 7 Summary of Profiles by MCAP Engagement and Knowledge Score

passive indicators. In the active indicators, students belonging to this profile showed above average interactions with videos and animations, average interactions with simulations, and average values in the matching tasks. We called this profile *Low Metacognition*.

The third profile was very small ($n=4$, 2%) and was characterized by average values in the active indicators, except very low values in the matching tasks, above-aver-

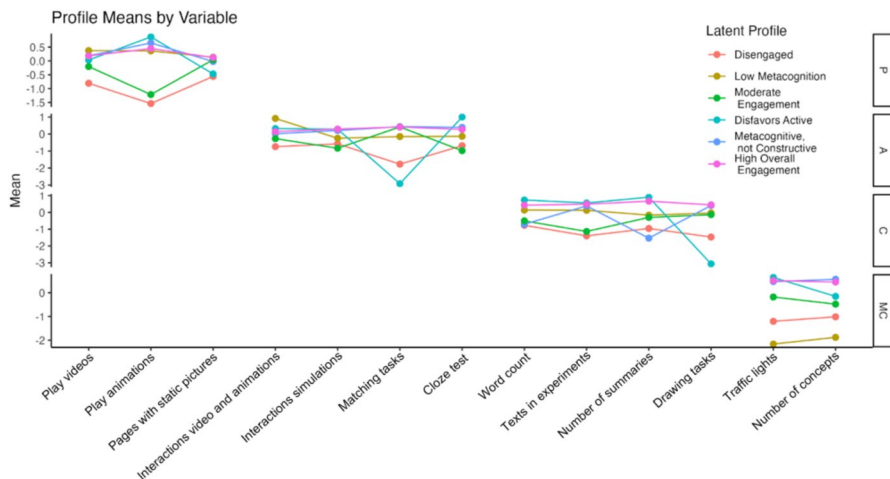


Fig. 8 Item Means by Profile

age values in the constructive indicators, except very low values in drawing tasks. From the passive indicators, this profile had above average values on interactions with animations, below-average time spent on pages with static pictures, and average interactions with videos. With respect to the metacognitive indicators, students in this profile had above average use of the traffic lights and slightly below-average values in the number of concepts. Therefore, we called this profile *Disfavors Active*.

The fourth profile ($n=29$, 12%) was characterized by average (playing videos, pages with static pictures) and above average (interactions with animations) values in the passive indicators, and average (interactions with videos, animations, and simulations) and above average (matching tasks) values in the active indicators. In the constructive indicators, this profile varied on the different tasks. Specifically, the fourth profile showed average values on drawing tasks, the cloze (fill-in-the-blank) test, and the number of texts written in experiments, but low number of written summaries and below average mean word count. The profile had high values on both metacognitive indicators. Therefore, we called this profile *Metacognitive but not Constructive*.

The fifth profile ($n=23$, 10%) was characterized by average (interactions with videos, time spent on pages with static pictures) and below average (interactions with animations) values in the passive indicators. In the active indicators, this profile had below average (interactions with simulations, cloze test), average (interactions with animations and videos), and above average (matching tasks) values in the active indicators. In the constructive indicators, the profile had average (drawing tasks, number of summaries written, mean word count) and below average (texts in experiments) values in the indicators. This profile had average frequency of using the traffic lights and below average number of concepts in the metacognitive indicators. Therefore, we called this profile *Moderate Engagement*.

The sixth and largest profile ($n=127$, 54%) was characterized by average (interactions with videos, time spent on pages with static pictures) and above average (interactions with animations) values in the passive indicators. In the active indicators, this profile had above average interactions with simulations and matching tasks and

average interactions with videos and animations. All constructive and metacognitive indicators had above average values. These students tended to complete all the activities available to them in the eChemBook environment, therefore, we called this profile *High Overall Engagement*.

Latent Profiles and Immediate Knowledge

We built two SEM models to predict the immediate knowledge test latent variable, one with the latent profiles as predictors of this outcome, and one model without (the null model). The null model only included prior knowledge as a covariate predicting achievement, thus we call it the “prior-knowledge-only model.” To answer RQ2, we then compared the fit of the more complex model against this prior-knowledge-only model using a chi-square test to understand whether the latent profiles explained variance in achievement outcomes. We answer RQ3 by comparing the results of the different profiles using post-hoc follow-up comparisons.

Omnibus Test

To test whether the inclusion of the profiles as predictors in our models significantly improved model fit, we tested two models. In the prior-knowledge-only model the regression coefficients from the dummy-coded indicators of profile membership predicting the immediate knowledge test were constrained to zero. As expected, this model did not fit well, $\chi^2(17)=54.26$, $p<.001$ (RMSEA=0.097, CFI=0.90, TLI=0.87, SRMR=0.084). In the prior knowledge + profiles model, these constraints were removed, thus allowing the immediate knowledge test to be predicted by profile membership. Overall, the model exhibited a near-perfect fit, $\chi^2(12)=8.97$, $p=0.71$ (RMSEA<.001, CFI=1, TLI=1, SRMR=0.017). We note, however, that these fit statistics might be expected given the model’s simplicity, which included only a single latent variable (the knowledge test) with three indicators. As shown in Table 2, a comparison of these models indicated that the latent profiles significantly predicted the immediate knowledge test outcome, $\Delta\chi^2(5)=33.71$, $p<.001$. This finding was further supported by comparison of AIC and BIC values between the models with and without profiles as predictors. The inclusion of the latent profiles explained 17% of the variance in the immediate knowledge test, $\Delta R^2=0.17$.

Table 2 Immediate Test Model Comparison

Models	R ² Knowledge	df	AIC	BIC	χ^2	diff	p
Prior Knowledge + Profiles	0.37	12	2286.0	2431.3	8.97		
Prior Knowledge Only	0.20	17	2321.2	2449.3	54.26	33.71	<.001

Note. The Chi-Squared Difference column is not directly calculable from the χ^2 column, as the difference is scaled per Satorra and Bentler’s (2001) robust Scaled Chi-Squared Difference Test procedure. The R² Knowledge column indicates the proportion of variance of the latent knowledge variable explained by the predictors

Post-Hoc Comparisons

To compare the individual latent profiles, we conducted a series of post-hoc tests. We report uncorrected p -values but interpret them using the Bonferroni correction for multiple comparisons ($\alpha=.005$). We calculated effect sizes by comparing the model-estimated standardized latent knowledge scores between conditions, resulting in an effect size comparable to a Cohen's d on the latent scale. We excluded the *Disfavors Active* profile from the post-hoc comparisons, as it only includes 4 students. As shown in Fig. 9A, the *High Overall Engagement* immediate knowledge test was significantly higher than *Disengaged* ($p<.005$, $d=0.87$) and *Moderate Engagement* ($p<.005$, $d=0.98$). The *Metacognitive, but not Constructive* profile was essentially the same distance from the other profiles, and significantly higher than the *Disengaged* ($p<.005$, $d=0.87$) and *Moderate Engagement* ($p<.005$, $d=0.98$). These effects were medium-to-large under Cohen's (1988) guidelines.

Regarding comparisons that did not meet our stricter Bonferroni criterion, the *High Overall Engagement* and *Metacognitive, but not Constructive* profiles also performed better than the *Low Metacognition* profile ($p=.03$ and $p=.02$, respectively). However, the *High Overall Engagement* and *Metacognitive, but not Constructive* profiles were not significantly different from one-another. The other three relatively low-performing profiles did not differ significantly.

Latent Profiles and Delayed Knowledge

We followed the same approach as for immediate knowledge to predict delayed knowledge.

Omnibus Test

In the prior-knowledge-only model, the regression coefficients from the dummy-coded indicators of profile membership predicting the delayed knowledge test were constrained to zero. As expected, this model did not fit well, $\chi^2(17)=62.24$, $p<.001$

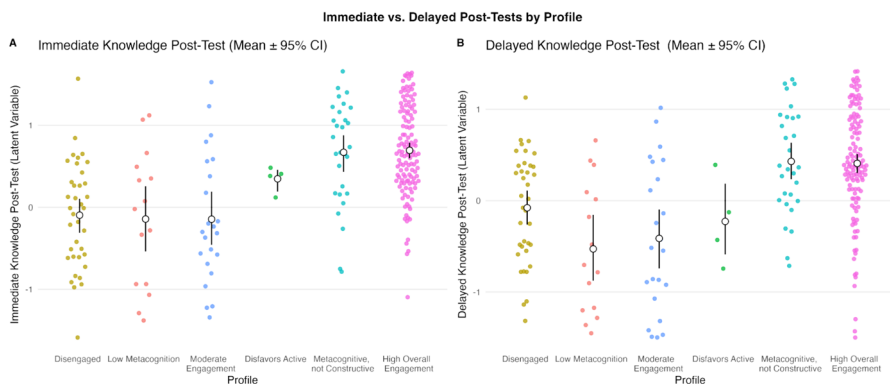


Fig. 9 Immediate (A) and Delayed (B) Knowledge Post-Hoc Comparisons and 95% Confidence Intervals

(RMSEA=0.11, CFI=0.86, TLI=0.83, SRMR=0.086). In the prior knowledge + profiles model, these constraints were removed, thus allowing the delayed outcome test to be predicted by profile membership. This model fit better, $\chi^2(12)=21.25$, $p=0.05$ (RMSEA=0.06, CFI=0.97, TLI=0.95, SRMR=0.028). A comparison of these models indicated that allowing the latent knowledge variable to be predicted by latent profile membership significantly improved model fit as assessed by AIC, BIC and a robust chi-squared difference test (see Table 3). In other words, we found that latent profiles significantly predicted the delayed knowledge test outcome, $\Delta\chi^2(5)=37.79$, $p<.001$. The inclusion of the latent profiles explained approximately 16% of the variance in the delayed knowledge test, $\Delta R^2=0.16$.

The Chi-Squared Difference column is not directly calculable from the χ^2 column, as the difference is scaled per Satorra and Bentler's (2001) robust Scaled Chi-Squared Difference Test procedure. The R^2 Knowledge column indicates the proportion of variance of the latent knowledge variable explained by the predictors.

Post-Hoc Comparisons

Again, as with the immediate knowledge outcome, we conducted a series of post-hoc tests. We applied the Bonferroni correction for multiple comparisons ($\alpha=.005$). We replicated the same general pattern of results as the immediate knowledge post-hoc tests, however in the delayed test, some comparisons shifted above or below our criterion of $p<.005$. As shown in Fig. 9B, the *High Overall Engagement* was significantly higher performing than the *Low Metacognition* profile ($p=.003$, $d=1.11$) and *Moderate Engagement* profiles ($p=.002$, $d=0.98$), but not the *Disengaged* profile ($p=.006$, $d=0.50$) after Bonferroni correction. Likewise, the *Metacognitive, but not Constructive* achieved significantly higher latent knowledge scores on the delayed test compared to the *Low Metacognition* profile ($p=.002$, $d=1.17$), but not the *Moderate Engagement* ($p=.006$, $d=1.04$) or *Disengaged* profiles ($p=.02$, $d=0.56$) after Bonferroni correction. Furthermore, the *High Overall Engagement* and *Metacognitive, but not Constructive* profiles were not significantly different from one-another. The other three relatively low-performing profiles also did not differ significantly when compared amongst themselves.

Pre-Existing Differences in Achievement and Motivation by Profile

One possible confound in our analysis relates to pre-existing differences in achievement or motivation between the profiles. For example, perhaps the *Disengaged* profile already had low prior knowledge or motivation prior to their engagement with the eChemBook, which could be the true cause of post-test achievement disparities. Although we account for this potential confound in our previous analyses (see Tables 2 and 3), we further assessed whether the profiles differed in motivation and prior knowledge.

Table 3 Delayed Test Model Comparison

Models	R^2 Knowledge	df	AIC	BIC	χ^2	diff	p
Prior Knowledge + Profiles	0.31	12	2404.5	2549.8	21.25		
Prior Knowledge Only	0.15	17	2435.5	2563.5	62.24	37.79	<.001

We found no significant differences in prior knowledge in an one-way ANOVA with conceptual knowledge pre-test as an outcome measure and profile membership as predictor, $F(5, 202)=0.99, p=.43$. To test for differences in motivation, we fit an omnibus model predicting domain-specific academic self-concept and interest in chemistry from the six latent profiles and compared it to a model wherein these latent outcome variables were not allowed to differ by profile membership. We found no significant differences in model fit between these two models, $\chi^2(10)=10.56, p=.39$. Thus, we found no evidence that students in different profiles differed in prior knowledge or motivation.

Discussion

The present study investigated how students' self-regulated engagement with different learning activities during learning with a digital chemistry textbook relate to immediate and delayed chemistry knowledge. Indicators of cognitive and metacognitive engagement were obtained from log files that captured the frequency and duration of students' learning activities. These indicators were categorized according to the ICAP framework (Chi & Wylie, 2014) as reflecting either passive (e.g., watching videos), active (e.g., interacting with a simulation), or constructive (e.g., drawing tasks) engagement. At the end of each learning unit, students could, but were not forced to, make judgments of learning by selecting the number of concepts from the unit they could explain and how well they could explain them. The frequency of making these judgments of learning was used as a measurement of metacognitive engagement. To better understand how students differ in their modes of cognitive and metacognitive engagement, we identified distinct engagement profiles and related them to immediate and delayed knowledge.

Profiles of Cognitive and Metacognitive Engagement (RQ1)

We identified six distinct profiles reflecting different combinations of cognitive and metacognitive engagement. Three profiles were characterized by consistently high (*High Overall Engagement*), low (*Disengaged*), and moderate (*Moderate Engagement*) engagement with (almost) all learning activities. The *High Overall Engagement* profile was the largest, suggesting that the majority of students completed the eChemBook diligently by working with most of the learning activities. The other three profiles showed distinct combinations of cognitive and metacognitive engagement. The *Low Metacognition* profile demonstrated a clear disfavor of metacognitive activities, whereas the *Metacognitive, but not Constructive* profile showed a preference for metacognitive activities but not for constructive activities. The last profile (*Disfavors Active*) showed a lack of engagement with active and constructive activities. We note that few students fell into this category; however, Spurk et al. (2020) recommend to retain profiles larger than 1% of the sample. Therefore, in addition to typically identified profiles showing consistently high, average, and low values on all activities (e.g., Wong et al., 2024), we identified profiles that demonstrate clear preference for specific types of engagement.

Identifying such profiles is beneficial for better understanding how individual differences in students' approaches to learning the same material relate to knowledge

acquisition. In particular, open-ended digital learning environments like the eChem-Book provide several types of activities and learning materials, which aim to foster different modes of cognitive and metacognitive engagement. Students can decide how much they want to engage with the different activities in a self-regulated manner and this information can be used to infer whether some modes of engagement (and their combination) are more beneficial for learning. Therefore, we compared the effectiveness of these activities for predicting immediate and delayed knowledge by classifying them based on the ICAP framework. As a next step, we used profile membership to predict immediate and delayed knowledge and explored differences in knowledge between the profiles.

Profile Membership in Relation to Immediate and Delayed Knowledge (RQ2)

Including the profiles in addition to prior knowledge, improved model fit and explained additional variance in both immediate and delayed knowledge. These results indicate that it is beneficial to not only consider students' starting point (i.e., their prior knowledge) but also their engagement with the platform when predicting learning outcomes. The predictive ability of latent profiles based on the ICAP framework to explain knowledge are in line with prior research using self-report measures to predict achievement (Wong et al., 2024). Next, we compared the profiles on immediate and delayed knowledge tests to further investigate whether some types of engagement are related to better learning outcomes than others.

Modes of Cognitive and Metacognitive Engagement and Knowledge (RQ3)

The third research question investigated whether some combinations of engagement are related to higher immediate and delayed knowledge. Although our analytical approach does not directly compare the different modes of engagement to each other, the results can be seen as a (partial) empirical test of the ICAP hypothesis that a constructive mode of engagement results in higher learning outcomes than active and passive (Chi & Wylie, 2014). Furthermore, our findings extend the ICAP framework by showing that the largest differences in both immediate and delayed knowledge are associated with differences in metacognitive engagement. In line with the ICAP framework, differences in passive and active engagement were less predictive of immediate and delayed knowledge.

Profiles characterized by high constructive and metacognitive (*High Overall Engagement*) or only metacognitive engagement (*Metacognitive but not Constructive*) had significantly higher immediate knowledge than profiles characterized by low metacognitive and constructive engagement (*Disengaged, Moderate Engagement*). In delayed knowledge, our findings showed that *High Overall Engagement* and *Metacognitive but not Constructive* had higher knowledge than the *Low Metacognition* profile and the *High Overall Engagement* profile also had higher knowledge than *Moderate Engagement*. The *Low Metacognition* profile had very low metacognitive engagement and focused mostly on passive and active activities.

In line with the ICAP framework, high constructive engagement (*High Overall Engagement*) was related to significantly higher immediate and delayed knowledge than low constructive engagement (*Disengaged, Moderate Engagement*). Therefore, our findings corroborate — at least correlationally — the assumption that engaging in constructive and generative activities is positively related to immediate and delayed learning outcomes (Fiorella & Kuhlmann, 2020; Fiorella & Mayer, 2016). While it has been proposed that constructive (and generative) activities will result in lasting knowledge, there are very few studies that have tested this assumption and they have only measured delayed knowledge after one week (e.g., Fiorella & Kuhlmann, 2020; Schmidgall et al., 2019). An important contribution of the present study is providing evidence for the role of constructive activities using delayed measurements of knowledge eight weeks later.

Importantly, the largest differences in the modes of engagement between profiles showing high learning outcomes (*High Overall Engagement* and *Metacognitive but not Constructive*) and *Low Metacognition* are in metacognitive engagement. This finding highlights the importance of metacognitive engagement for both immediate and delayed knowledge. Metacognitive engagement was measured by the frequency of making judgments of learning and the number of concepts understood. Students were not required to make these metacognitive judgments and our findings show that more frequent metacognitive monitoring was related to the highest learning outcomes. In line with metacognition frameworks (e.g., Nelson & Narens, 1990), engaging in monitoring may have triggered appropriate control processes and resulting in students allocating their study time better and focusing on concepts that required further learning (Ackerman, 2014; Janes et al., 2018). Furthermore, our findings on the importance of metacognitive engagement for immediate and delayed learning provide further evidence for the importance of self-regulated learning during computer-supported learning (Brady et al., 2022; Scheiter et al., 2009; Winters et al., 2008).

Another interesting element of our findings relates to the relative *unimportance* of constructive activities. Our findings demonstrate that, partially in line with the ICAP framework, high engagement was related to high immediate and delayed knowledge. However, as evidenced by the performance of the *Metacognitive, but not Constructive profile*, we do not find that constructive engagement was absolutely necessary for high achievement. In fact, the performance of the *Metacognitive, but not Constructive profile* was not statistically distinguishable from the *High Overall Engagement profile*. A potential explanation for this finding could be what Koedinger et al. (2016) called the *doer effect*, wherein generally the more active practice (including both active and constructive forms of practice), the better students do. This finding has been replicated across several online and offline learning environments (Butler et al., 2025; Van Campenhout et al., 2021) Indeed, we find that high amounts of engagement predict better learning as the *doer effect* literature predicts. Yet, we find less striking correlations between constructive engagement and achievement than the ICAP framework might have led us to believe.

Theoretical and Practical Implications

The present study empirically tested the relative importance of cognitive and metacognitive engagement, obtained from students' actions within a digital textbook, for immediate and delayed knowledge. The ICAP framework and the derived hypotheses (Chi & Wylie, 2014; Chi et al., 2018) were recently criticized for lacking empirical testing in ecologically valid studies or only investigating one mode of engagement at a time (Thurn et al., 2023). One of the strengths of this study is empirically testing the (I)CAP hypothesis in real chemistry classrooms including two learning sessions and measurements of both immediate and delayed knowledge. Furthermore, while we do not investigate all four modes of engagement, the present study considers the three modes of engagement that are relevant to individual learning settings. Furthermore, our findings on the importance of metacognitive engagement suggest that the ICAP framework could be extended to include metacognitive engagement as an additional dimension (ICAP-M). The results of the present study demonstrate the role of metacognitive engagement, in addition to cognitive engagement, in individual self-regulated learning settings. Given that interactive engagement is posited as the most effective type of engagement for learning (Chi & Wylie, 2014), it is important to further investigate the role of socially shared metacognitive regulation during collaborative learning settings.

The present study also contributes to the growing number of studies using the ICAP framework to classify students' engagement using different measures (e.g., self-report, Wong et al., 2024; classroom observations, Goldberg et al., 2021) by classifying engagement based on logfiles of student activity. Learning analytics research has been criticized for not relating logfile measurements of students' learning processes to learning outcomes or for identifying learning activities from log files without theoretical grounding (Motz et al., 2023; Wise & Shaffer, 2015). In this study, we used a theory-driven approach to classify log file data of student activity into different modes of cognitive and metacognitive engagement using multiple behavioral indicators. In particular, between two and five different indicators were used for each mode of engagement. Our findings on using logfile data for inferring different modes of cognitive and metacognitive engagement contribute to learning analytics research by linking the theoretically grounded classification of log files with immediate and delayed learning outcome measures. In particular, this study shows that the ICAP framework can be used to classify logfile measurement of engagement, in addition to self-reports and classroom observations.

Limitations

One limitation of this study is that we could only investigate and compare passive, active, and constructive modes of engagement. The ICAP framework postulates that interactive engagement is the form of engagement, associated with the highest learning outcomes (Chi & Wylie, 2014). Since the eChemBook was designed to support individual learning, we could not include data about interactive engagement. Nevertheless, the present study compared the contribution of three of the four modes of cognitive engagement simultaneously using multiple indicators for each mode. This limits our findings to individual learning settings and the role of metacognitive

engagement in collaborative learning settings requires further investigation. In future research, the relative importance of socially shared metacognitive regulation during tasks affording interactive engagement should be investigated. Additionally, the current dataset only included direct indicators of metacognitive monitoring and no other core elements of metacognition, such as metacognitive control or metacognitive knowledge (Dunlosky & Metcalfe, 2008).

A second limitation of our study is that for some indicators of cognitive engagement, we focused on the presence or absence of engaging in a particular activity and not the quality of the resulting student products. For example, the indicator “texts in experiments” provides information on whether students wrote something in the text box. We account for the length of these texts, as a proxy for engagement, and thus quality. Therefore, our measurement of quantity, instead of quality, of constructive engagement might have underestimated the impact of constructive activities. In future research, it is important to also measure the quality and depth of engagement. Likewise, the reliability of our prior knowledge scale was low, possibly due to students having different (and possibly unrelated) misconceptions about the particle model of matter prior to learning. The reliability of the same test was much higher in the immediate and delayed knowledge measurements.

Third, the observational study design prevents us from establishing a causal relationship between the modes of engagement and knowledge. In future research, an experimental manipulation of the modes of engagement in relation to learning outcomes is needed. As acknowledged by Bray and Dziak (2018), latent profile analysis often resembles “an art and a science” (p. 107). We do not necessarily claim that our exact number or configuration of profiles would generalize to other learning situations, though we might expect general theoretically-supported themes to be relatively stable across contexts (e.g., some students exhibit higher and lower engagement). Due to its small size, we could not investigate learning outcomes for the *Disfavors Active* profile and a replication of our profile solution with a larger sample size is recommended.

Lastly, we acknowledge our sample was drawn from two German states, which limits generalizability. The majority of the sample were in the *High Overall Engagement* profile, suggesting that a substantial proportion of the students essentially completed all available activities. However, we note that this level of engagement could be particular to the context or the sample of this study, whereas other educational environments might not lead to such sustained student engagement. Future research could expand upon the limits and boundary conditions of our findings by providing ample practice opportunities for students, requiring them to make tough decisions about which activities to engage in, and which activities *not to engage in*.

Conclusion

This study used a theoretically-informed analysis of student logfile data in an ecologically valid context to better understand which modes of cognitive and metacognitive engagement are related to higher chemistry knowledge. The immediate and delayed measurements of knowledge showed relationships with metacognitive engagement, even when controlling for prior knowledge. Particularly, the profiles *Metacognitive*, *but*

not Constructive and *High Overall Engagement* performed equally well, which suggests that metacognitive engagement is an additional important factor for learning in digital learning environments. Therefore, we propose metacognitive engagement as an important extension of the ICAP framework (ICAP-M), particularly in individual, self-regulated learning settings. Our findings demonstrate the importance of metacognition in addition to cognitive engagement and provide important implications for educational practice and the design of digital learning materials. First, it might be beneficial for teachers to combine formative assessments with metacognitive judgments to improve students' monitoring accuracy. Second, digital materials should include tasks that promote not only active and constructive engagement, but deliberately foster metacognitive engagement.

Appendix

Table 4 The Number and Type of Multimedia Elements within Each Unit of the eChemBook

	Motivational Teaser	Basic Text	Video Experiment	Interactive Exercises	Summary
Unit 1	113 words 1 picture	5 pages, 927 words, 7 pictures, 1 animation	2 pages, 196 words, 1 video, 4 pictures	2 exercises, 50 words, 4 pictures	1 page, 128 words, 1 picture, 1 traffic light
Unit 2	86 words, 1 picture	4 pages, 397 words, 4 pictures	2 pages, 188 words, 1 video, 1 picture	2 exercises, 50 words, 2 pictures	1 page, 20 words, 1 picture, 1 traffic light
Unit 3	91 words, 1 picture	3 pages, 428 words, 2 animations, 2 simulations, 1 cloze test	2 pages, 135 words, 1 video, 2 pictures	2 exercises, 60 words, 4 pictures	1 page, 133 words, 3 pictures, 1 traffic light
Unit 4	62 words, 1 picture	5 pages, 658 words, 2 simulations, 1 video, 1 cloze test	2 pages, 163 words, 1 video, 2 pictures	2 exercises, 80 words, 1 video, 3 pictures	1 page, 131 words, 1 picture, 1 traffic light
Unit 5	74 words, 1 picture	5 pages, 658 words, 4 pictures, 3 animations, 1 simulation	2 pages, 100 words, 1 video, 1 picture	2 exercises, 40 words, 2 pictures	1 page, 102 words, 1 picture, 1 traffic light
Unit 6	49 words, 1 picture	7 pages, 910 words, 4 pictures, 1 animation	2 pages, 163 words, 1 video, 2 pictures	2 exercises, 55 words, 8 pictures	1 page, 163 words, 1 picture, 1 traffic light

Table 5 Descriptive Statistics (Mean or Percentage Correct, Standard Deviation, Range) of All Variables

Indicator	Mean or % Correct	SD	Range
Pre-test ^a	67.14	13.64	[26.67–100]
Domain academic self-concept	1.40	0.61	[0.2–3]
Domain-specific interest	1.79	0.69	[0–3]
Post-test ^a (misconceptions)	81.96	14.87	[33.33–100]
Post-test ^a (recall)	61.87	21.83	[3.45–100]
Post-test ^a (transfer)	32.66	17.47	[0–71.74]
Delayed post-test ^a (misconceptions)	81.99	15.19	[26.67–100]
Delayed post-test ^a (recall)	61.72	22.60	[3.45–100]
Delayed post-test ^a (transfer)	32.63	18.02	[0–80.43]
Play videos	15.22	8.16	[0–52]
Play animations	3.36	1.02	[0–5]
Pages with static pictures (minutes)	18.75	6.59	[0–37.51]
Interactions with animations and videos	42.84	27.02	[0–192]
Interactions in simulations	16.12	14.32	[0–72]
Matching tasks	1.74	0.51	[0–2]
Cloze (fill-in-the-blank) test	4.66	2.85	[0–10]
Texts in experiments	7.32	2.09	[0–9]
Number of summaries	3.93	2.02	[0–6]
Word count (average)	232.68	120.47	[0–633]
Drawing tasks	8.21	2.35	[0–10]
Traffic lights	4.63	1.73	[0–6]
Concept checkboxes	7.57	3.68	[0–12]

Note. See Table 1 for the definition of each indicator

^a The value represents the percentage of correct items from the total

Table 6 Model Comparison Table

Profiles	LL	AIC	AWE	BIC	CAIC	CLC	KIC	SABIC	ICL	Entropy	Smallest Profile Size	Largest Profile Size
1	-4336.3	8724.7	9032.6	8814.6	8840.6	8674.7	8753.7	8732.2	-8814.6	1	235	235
2	-3936.8	7953.5	8428.3	8091.9	8131.9	7875.4	7996.5	7965.1	-8096	0.97	62	173
3	-3820.5	7749.0	8390.6	7935.8	7989.8	7642.9	7806	7764.6	-7938.4	0.99	15	173
4	-3750.4	7636.8	8445.3	7872	7940	7502.7	7707.8	7656.5	-7879.3	0.97	18	137
5	-3676.4	7516.8	8492.2	7800.5	7882.5	7354.7	7601.8	7540.5	-7806.5	0.98	6	132
6	-3599.7	7391.4	8533.7	7723.5	7819.5	7201.3	7490.4	7419.2	-7729.7	0.98	4	127
7	-3560.9	7341.8	8650.9	7722.3	7832.3	7123.7	7454.8	7373.7	-7731.6	0.97	4	117
8	-3502.7	7253.4	8729.4	7682.4	7806.4	7007.4	7380.4	7289.4	-7691.4	0.98	4	117
9	-3481.0	7238.0	8880.9	7715.4	7853.4	6963.9	7379.0	7278.0	-7731.7	0.96	4	104
10	-3494.0	7291.9	9101.7	7817.8	7969.8	6989.8	7446.9	7336.0	-7836.4	0.96	3	102

Note. LL = log likelihood; AIC = Akaike information criterion; AWE = Approximate weight of evidence; BIC = Bayesian information criterion; CAIC = Consistent Akaike's information criterion; CLC = Classification likelihood criterion; KIC = Kullback information criterion; SABIC = Sample-adjusted BIC; ICL = Integrated complete-data likelihood criterion

Authors' contributions Yoana Omarchevska—Conceptualization, Methodology, Formal Analysis, Writing—original draft; Brendan Schuetze—Conceptualization, Methodology, Formal Analysis, Visualization, Writing—original draft; Juliane Richter—Conceptualization, Methodology, Investigation, Project Administration, Writing—review and editing; Sascha Schanze—Conceptualization, Funding Acquisition, Supervision, Writing—review and editing; Nina Ulrich—Material Development, Investigation, Project Administration; Katharina Scheiter—Conceptualization, Funding Acquisition, Supervision, Writing—review and editing.

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Data availability The datasets generated during and analyzed during the current study are available from the corresponding author upon reasonable request. The analysis code and output can be found at the following OSF link: https://osf.io/uypfk/?view_only=11d819491f7b4b158ef984ebf02f470e

Declarations

Ethics approval The research project received ethical approval from the Ministry of Education in Lower Saxony and the school administration in Baden-Württemberg.

Consent Written informed consent was obtained from the students and their parents prior to participation. Participation was voluntary.

Competing interests The authors have no relevant financial or non-financial interests to disclose.

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